

Rockchip RV1126B Linux6.1 SDK Release Note

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Preface

Overview

The document presents Rockchip RV1126B Linux6.1 SDK release notes, aiming to help engineers get started with RV1126B Linux6.1 SDK development and debugging faster.

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Chipset and System Support

Chipset	Buildroot	Debian	Yocto
RV1126B, RV1126B-P	Y	Y	Y

Revision History

Date	Version	Author	Revision History
2025-03-26	V0.0.1	Caesar Wang	Alpha version
2025-05-15	V0.1.0	Caesar Wang	Beta version
2025-06-20	V1.0.0	Caesar Wang	Release version
2025-09-20	V1.1.0	Caesar Wang	Release V1.1.0 version

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1. Overview

This SDK supports three systems based on Buildroot 2024.02, Debian 12, and Yocto 5.0, with the kernel based on Kernel 6.1 and the bootloader based on U-boot v2017.09. It is suitable for RV1126B EVB development boards and all Linux products developed based on this board.

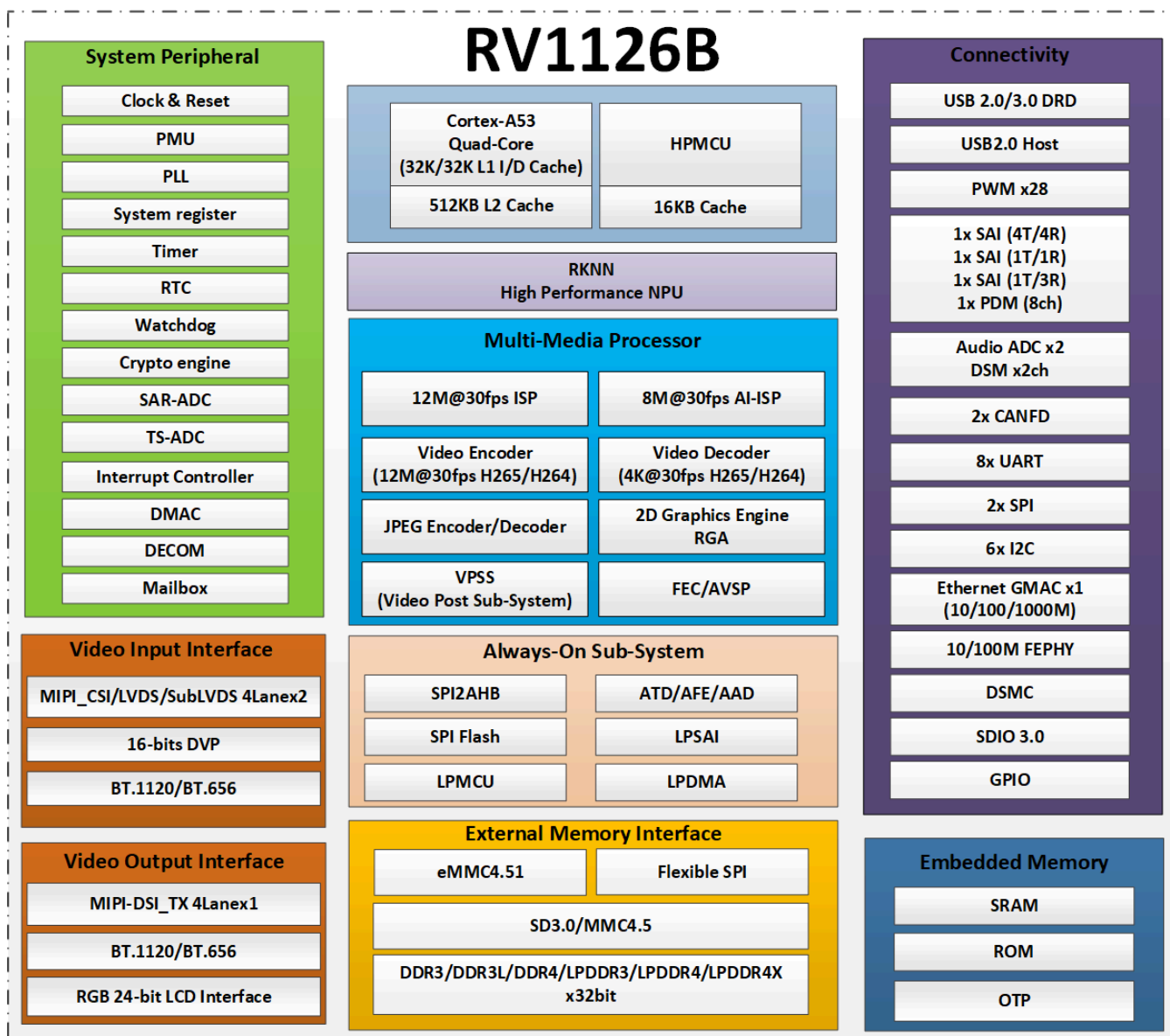
The development package is suitable for AIoT solutions in industries such as industrial applications, smart security, industrial vision, and robotic vacuum cleaners. It offers flexible data path combination interfaces to meet customers' customization needs. For specific function debugging and interface details, please refer to the documents in the project directory's docs/folder.

2. Main Functions

Function	Module Name
System	Buildroot, Debian, Yocto
Partition table	uboot, misc, boot, recovery, rootfs, oem, userdata
File System Type	EXT2/3/4, VFAT, NTFS, UBIFS, SquashFS
Upgrade Recovery	OTA, AB, Recovery
Secure Boot	SecureBoot
Stress Test Tool	ROCKCHIP_TEST
Data communication	Wi-Fi, Ethernet, USB, SDIO, DSMC, CANFD, UART, SPI, I2C interface
Applications	Multimedia Playback, Camera Preview, Desktop UI

2.1 Hardware functions

For specific hardware interface functions, please refer to the RV1126B chip diagram below



2.2 Chip Difference Explanation

The RV1126B is the full - featured version, including complete functions such as USB3.0, Audio Codec, and Ethernet PHY. The RV1126B-P is the Pin2Pin version of the RV1126, allowing customers who previously developed products with the RV1126 to directly replace it with the RV1126B-P without hardware redesign. For specific functional differences and compatibility details, please refer to the document "RV1126&RV1126B-P_Differences_Description_and_Compatibility_Design_Notes_V1.26_20250604_CN&EN.pdf".

2.2.1 Functional Differences

Function	RV1126	RV1126B-P (pin to pin)	RV1126B
Fast - booting	Supported	Supported	Supported
AOV	Supported	Supported	Supported
Pre - recording	N/A	N/A	Supported

2.2.2 Specification Differences

Specification	RV1126	RV1126B-P (pin to pin)	RV1126B
CPU	4*A7	4*A53	4*A53
Audio	SAI(I2S/PCM/TDM)+PDM+ 2*DSM(PWM Audio)	SAI(I2S/PCM/TDM)+PDM+ 2*DSM(PWM Audio)	SAI(I2S/PCM/TDM)+PDM+ 2DSM(PWM Audio)+2Audio ADC
SARADC	6	6	24
PWM	16	28	28
MIPI CSI	2x4lane	2x4lane	2x4lane/4x2lane
USB	USB2.0 DRD+USB2.0 HOST	USB2.0 DRD+USB2.0 HOST	USB2.0/3.0 DRD+USB2.0 HOST
RTC	N/A	N/A	Built - in
Ethernet	GMAC	GMAC	GMAC+FE PHY

3. How to Get the SDK

The SDK is distributed via Rockchip's official code repository and can only be downloaded with **authorized access**. For offline acquisition, you may request a **local source code package** (contact Rockchip's technical support team for assistance). For details on the compilation and development environment, refer to Chapter 4, Software Development Guide.

3.1 Download Authorization

If you have completed the account creating and need to generate your own HTTPS password before downloading the code, please download the repository `https://gerrit.rock-chips.com:8443/repo-release/tools/script` and download the authorization through the `script/Gernerate-Credential.sh` under the repository, this is shown below.

```
mkdir ~/rk_gerrit_password
cd ~/rk_gerrit_password
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/script
cd ./script
./Generate-Credential.x
```

The log of running the authorization script is as follows:

```
Adding configuration for Gerrit server to /home/Test123/rk_gerrit_password
*****
*****
*****
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chips.com
Please input your SSH private key filename (e.g. id_rsa):
Your Credential check has passed
New git credential will be added into your history!!
You are All Set!!
```


3.2 Get Source Code from Rockchip Code Server

To get RV1126B Linux6.1 SDK software package, customers need an account to access the source code repository provided by Rockchip. In order to be able to obtain code synchronization, please provide SSH public key for server authentication and authorization when apply for SDK from Rockchip technical window. About Rockchip server SSH public key authorization, please refer to Chapter 6 [SSH Public Key Operation Introduction](#).

RV1126B Linux6.1 SDK download command is as follows:

```
repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo \
-u https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b \
rv1126b -m rv1126b_linux6.1_release.xml
```

Repo, a tool built on Python script by Google to help manage git repositories, is mainly used to download and manage software repository of projects. The download address is as follows:

```
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/repo
```

3.3 Get Source Code from Local Compression Package

For quick access to SDK source code, Rockchip Technical Window usually provides corresponding version of SDK initial compression package. In this way, developers can get SDK source code through decompressing the initial compression package, which is the same as the one downloaded by repo.

Take RV1126B_LINUX6.1_SDK_RELEASE_V1.1.0_20250920.tgz as an example. After getting a initialization package, you can get source code by running the following command:

```
mkdir rv1126b
tar xvf RV1126B_LINUX6.1_SDK_RELEASE_V1.1.0_20250920.tgz -C rv1126b
cd rv1126b
.repo/repo/repo sync -l
.repo/repo/repo sync -c
```

Developers can update via `.repo/repo/repo sync -c` command according to update introductions that are regularly released by FAE window.

If you encounter issues downloading the repository, you can use the `--force-sync` parameter to force an update, as in `.repo/repo/repo sync -c --force-sync`. Before doing so, please ensure that any local modifications have been backed up.

After updating the SDK code, a clean operation is required. For example: `./build.sh cleanall`

Note:

The software release version can be viewed through engineering XML, as follows:

```
.repo/manifests$ realpath rv1126b_linux6.1_release.xml
e.g., the printed version is v1.1.0 and the update time is 20250920
<SDK>/.repo/manifests/release/rv1126b_linux6.1_release_v1.1.0_20250920.xml
```

4. Software Development Guide

For software development, please refer to the quick start documents in the project directory:

```
<SDK>/docs/en/RV1126B/Quick-start/Rockchip_RV1126B_Quick_Start_Linux_EN.pdf
```

5. Hardware Development Guide

For hardware development, please refer to the following documents in the project directory:

```
<SDK>/docs/en/Socs/RV1126B/Hardware$  
├─ RV1126&RV1126B-  
P_Differences_Description_and_Compatibility_Design_Notes_V1.26_20250604_CN&EN.pdf  
├─ RV1126&RV1126B_Function_Mapping_Table_V1.0_20250527.xlsx  
├─ RV1126B-P_Package_PinLength_V1.0_20250318_EN.xlsx  
├─ RV1126B-P_PinOut_V1.0_20250519_EN.xlsx  
├─ RV1126B_EVB1_User_Guide_V1.0_20250623_EN.pdf  
├─ RV1126B_Methods_for_Processing_Unused_Pins_V1.0_20250606_EN.xlsx  
├─ RV1126B_Package_PinLength_V1.0_20250318_EN.xlsx  
└─ RV1126B_PinOut_V1.1_20250519_EN.xlsx
```

6. SSH Public Key Operation Introduction

Please follow the introduction in the “Rockchip_User_Guide_SDK_Application_And_Synchronization_EN” to generate an SSH public key and send the email to sw.fae@rock-chips.com, to get the SDK code.

This document will be released to customers during the process of applying for permission.

6.1 Key Authority Management

Server can monitor download times and IP information of a key in real time. If an abnormality is found, download permission of the corresponding key will be disabled.

Keep the private key file properly. Do not grant second authorization to third parties.

6.2 Reference Documents

For more details, please refer to document

“/docs/en/Others/Rockchip_User_Guide_SDK_Application_And_Synchronization_EN.pdf”